

SHANKA SUBHRA MONDAL

shankasubhra@gmail.com; <https://shanka123.github.io/>

EDUCATION

Princeton University	<i>2019 - 2025</i>
Ph.D. in Electrical & Computer Engineering	GPA: 3.91/4
Indian Institute of Technology, Kharagpur	<i>2015 - 2019</i>
B.Tech. in Electronics & Electrical Communications Engineering	GPA: 9.51/10

WORK EXPERIENCE

Accenture Center for Advanced AI, Mountain View, California *Jan 2026 - Present*
Senior Research Scientist

- Leading a research project on joint multi-agent reinforcement learning with visual grounding in flow based VLAs for robotics applications.
- Led a research project to develop a modular multimodal GUI agent. Jointly fine-tuned two 7B VLMs, one serving as a task decomposer and the other as an actor, using a hierarchical extension of GRPO. Achieved state-of-the-art performance among multimodal web agents. (**Preprint under review**)
- Building AI optimization assets focused on prompt optimization and adaptive model routing.

Microsoft Research, New York *Summers 2023 and 2024*

Ph.D. Research Intern | Mentor: Ida Momennejad

Improved multi-step planning in LLMs. (**Nature Communications 2025**)

- Developed a novel black-box, modular, multi-agent architecture using LLMs, inspired by specialized prefrontal cortex modules in the human brain.
- Fine-tuned GPT-4 via the Microsoft Azure API to serve as specialized modules (e.g., detecting errors and determining whether a goal has been achieved).
- Studied transfer generalization across multi-step reasoning tasks in different LLM-based reasoning agents.
- Research ideas were incorporated into Microsoft products.
- Collaborated with the team in the office of Microsoft's CTO.

Adobe Research, Bangalore, India *Summer 2018*

Research Intern | Mentors: Subrata Mitra, Ramanuja Simha

Developed a scheduler based on deep reinforcement learning to optimize resource utilization and application performance. (**AAAI 2021**)

RESEARCH EXPERIENCE

Princeton University *2019 - 2025*

Graduate Researcher, Princeton Neuroscience Institute | Ph.D. Advisor: Jonathan D. Cohen

My Ph.D. work used insights from cognitive science and neuroscience to design neural networks with human-like systematic and sample-efficient generalization.

- Incorporated recurrent processing with memory tokens into large vision-language models.
- Integrated relational inductive biases with object-centric representations for abstract visual reasoning tasks. Developed novel architectures using slot attention, pairwise relational operators, transformers, and Abstractors. (**NeurIPS 2023, ICML 2024**)

- Developed a novel architecture for abstract visual reasoning that combines object-centric representations from slot attention with a transformer-based reasoning module. (**ICLR 2023**)
- Developed an attention mechanism inspired by determinantal point processes over biologically inspired structured embeddings (grid cells) of sensory information. (**eLife 2023**)

Princeton University

2019 - 2021

Graduate Researcher | Advisor: Tom Griffiths

Developed neural network models with improved generalization and robustness to adversarial attacks.

Princeton University

2019 - 2021

Graduate Researcher | Advisor: Sebastian Seung

Developed deep learning models for semantic segmentation in electron microscopy images, 3D point clouds, and meshes of neuroscience data.

Indian Institute of Technology, Kharagpur

2018 - 2019

Undergraduate Researcher | Advisor: Debdoot Sheet

Developed a multi-task deep learning model to predict tool and phase information in surgical videos.

Indian Statistical Institute, Kolkata

Summer 2017

Summer Intern, Machine Intelligence Unit | Advisor: Sushmita Mitra

Implemented SpikeProp and spike-timing-dependent plasticity algorithms for neural networks.

PUBLICATIONS

Webb, T.W.*, **Mondal, S.S.***, Momennejad, I. (2025). A brain-inspired agentic architecture to improve planning with LLMs. *Nature Communications*. <https://doi.org/10.1038/s41467-025-63804-5>

Mondal, S.S., Cohen, J.D., Webb, T.W. (2024). Slot Abstractors: Toward Scalable Abstract Visual Reasoning. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.

Mondal, S.S.*, Webb, T.W.*, Cohen, J.D. (2023). Systematic Visual Reasoning through Object-Centric Relational Abstraction. *Advances in Neural Information Processing Systems (NeurIPS)*.

Mondal, S.S., Frankland, S.M., Webb, T.W., Cohen, J.D. (2023). Determinantal point process attention over grid cell code supports out of distribution generalization. *eLife*. <https://doi.org/10.7554/eLife.89911.3>

Mondal, S.S.*, Webb, T.W.*, Cohen, J.D. (2023). Learning to reason over visual objects. In *Proceedings of the 11th International Conference on Learning Representations (ICLR)*.

Mondal, S.S.*, Sheoran, N.*, Mitra, S. (2021). Scheduling of time-varying workloads using reinforcement learning. In *Proceedings of the 35th AAAI Conference on Artificial Intelligence*.

Mitra, S., Sheoran, N., Simha, R., **Mondal, S.S.**, Dhake, N., Nehra, R. (2020). Self-learning scheduler for application orchestration on shared compute cluster. *US Patent*.

Mitra, S., **Mondal, S.S.**, Sheoran, N., Dhake, N., Nehra, R., Simha, R. (2019). Deepplace: Learning to place applications in multi-tenant clusters. In *Proceedings of the 10th ACM SIGOPS Asia-Pacific Workshop on Systems*.

Nandy, A., **Mondal, S.S.** (2019). Kinship verification using deep siamese convolutional neural network. In *Proceedings of the 14th IEEE International Conference on Automatic Face and Gesture Recognition (FG) Workshop*.

Mondal, S.S., Sathish, R., Sheet, D. (2018). Multitask learning of temporal connectionism in convolutional networks using a joint distribution loss function to simultaneously identify tools and phase in surgical videos. In *Proceedings of the 5th MedImage Workshop at the 11th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*.

CONFERENCE AND WORKSHOP PRESENTATIONS

- Slot Abstractors: Toward Scalable Abstract Visual Reasoning. Poster presented at the 41st International Conference on Machine Learning (ICML), Vienna, Austria, 2024.
- Object-Centric Abstract Reasoning. Talk presented at the Workshop on Abstraction and Relational Reasoning, Yale University, 2023.
- Systematic Visual Reasoning through Object-Centric Relational Abstraction. Talk recorded and poster presented at the 37th Conference on Neural Information Processing Systems (NeurIPS), New Orleans, 2023.
- Learning to reason over visual objects. Talk recorded and poster presented at the 11th International Conference on Learning Representations (ICLR), Kigali, Rwanda, 2023.
- Scheduling of time-varying workloads using reinforcement learning. Talk recorded and poster presented at the 35th AAAI Conference on Artificial Intelligence, 2021.
- Multitask learning of temporal connectionism in convolutional networks using a joint distribution loss function to simultaneously identify tools and phase in surgical videos. Talk and poster presented at the 5th MedImage Workshop at the 11th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP), IIIT Hyderabad, 2018.
- Surgical Workflow Analysis Challenge. Presented the winning solution at the International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2018.
- Investment Ranking Challenge. Presented the third-place solution at the IEEE Data Science Workshop, EPFL, Switzerland, 2018.

AWARDS AND HONORS

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|---|--------------------|
| • NeurIPS Scholar Award | <i>2023</i> |
| • Karl Storz Endoscopic Workflow Award, MICCAI | <i>2018</i> |
| • Goralal Syngal Memorial Scholarship, IIT Kharagpur | <i>2017 - 2018</i> |

SERVICE

- **Teaching Assistant**, Princeton University, Department of Statistics and Machine Learning
Modern Data Science Methods, Spring 2023 and Spring 2024
- **Reviewer** (NeurIPS, ICLR)

SKILLS AND KNOWLEDGE

Courses: Computer Vision, Large-Scale Data Optimization, Biomedical Imaging, Mathematical Foundations of Reinforcement Learning, Matrix Algebra, Probability and Stochastic Processes, Programming and Data Structures, Algorithms, Network Theory, Intelligent Game Design, Signals and Systems, Econometric Analysis, Digital Signal Processing, Computer Architecture, Operating Systems, Digital Image Processing, Neural Networks, Artificial Intelligence, Machine Intelligence and Expert Systems, Deep Learning, Advanced Machine Learning, Pattern Recognition.

Programming Languages: Python, C/C++, MATLAB, C#, Java, Verilog HDL.

Software & Tools: PyTorch, DeepSpeed, TensorFlow, Keras, MXNet, scikit-learn, Matplotlib, Xilinx, Simulink, Arduino, Unity 3D, PSpice, Android Studio, SolidWorks.